

EXECUTIVE BRIEF · VERSION 2

ProposalPilot

Applied Fixes · Corrected Findings · Owner Action Items · Billion Dollar Build Strategy

Updated May 3, 2026 · Prepared by Perplexity Computer · Confidential

Repository: github.com/Ryan-Richmond/RFPPProject · Database: Supabase ProposalPilot · Live:
proposal-pilot-vert.vercel.app

SECTION 1

Corrections to Version 1 Findings

The first review was conducted using a cloud browser without access to the authenticated application. Three findings were incorrect and are retracted here. All other findings from v1 remain valid.

RETRACTED	Embedding Dimension Mismatch The brief incorrectly stated that migration 001 creates vector(768). Direct SQL query confirmed atttypmod=1024 — the column is already vector(1024) and correctly aligned with the Perplexity embeddings model. No fix required.
RETRACTED	Missing match_evidence_chunks RPC The function does exist. A direct pg_proc query confirmed it is defined, uses vector(1024) in its cast, and is stable. It was already correctly implemented. No fix required.
RETRACTED	Live Deployment Missing The GovCon ProposalPilot application is fully deployed at proposal-pilot-vert.vercel.app. The browser test hit a different unrelated app at proposal-pilot.vercel.app (no "-vert"). The live product includes all modules: Workspace, Opportunities, Knowledge Base, RFP Analysis, Drafting, Compliance, Team, Profile, and Computer Ops.
CLARIFIED	AI Provider Conflict (Gemini vs. Perplexity) The Gemini integration is intentional — Gemini Flash is a low-cost QA/testing provider. All 5 core services currently import from lib/ai/gemini, which is the active QA client. Perplexity is the designated production provider. The .env.local.example correctly documents PERPLEXITY_API_KEY as the sole production key. The action item is an environment-controlled provider switch, not a code conflict.

SECTION 2

Applied Database Fixes

Two migrations were applied directly to the ProposalPilot Supabase database (project ID: fzukcuiwiexrfquuzqii) on May 3, 2026. All changes were verified with follow-up SQL queries confirming successful application.

Migration 020 — Performance Indexes

Status: APPLIED ✓ — Verified applied. 7 new indexes created across 5 tables.

INDEX	TABLE	TYPE	PURPOSE
idx_solicitations_workspace_status	solicitations	Composite	Covers most frequent dashboard filter: workspace + status
idx_proposal_drafts_workspace_statuses	proposal_drafts	Composite	Proposal list views scoped to workspace + status
idx_opportunities_workspace_status	opportunities	Composite	Opportunity pipeline board filters
idx_evidence_chunks_naics_codes	evidence_chunks	GIN	Array containment queries for NAICS code matching on evidence
idx_evidence_chunks_keywords	evidence_chunks	GIN	Keyword array search on indexed evidence chunks
idx_agent_operations_workspace_created	agent_operations	Composite	Computer Ops dashboard timeline queries
idx_audit_logs_workspace_created	audit_logs	Composite	Audit log queries scoped to workspace chronologically

Migration 021 — Security Hardening

Status: APPLIED ✓ — Verified: all 9 functions now show proconfig=["search_path=public, extensions"]. RLS policies on workspaces and workspace_members updated to use (SELECT auth.uid()).

FIX	DETAIL	IMPACT
Mutable search_path fixed on 9 functions	Added SET search_path = public, extensions to: handle_updated_at, handle_new_workspace, is_workspace_admin, is_workspace_member, validate_workspace_invite, storage_object_workspace_id, handle_new_user_workspace_invite, redeem_workspace_invite, match_evidence_chunks	Eliminates schema injection risk on all public-facing DB functions

RLS re-evaluation fixed on workspaces	workspace_owner_all policy now uses (SELECT auth.uid()) instead of auth.uid()	Prevents per-row auth.uid() re-evaluation — improves RLS scan performance at scale
RLS re-evaluation fixed on workspace_members	workspace_members_insert policy now uses (SELECT auth.uid())	Same performance improvement on membership checks

SECTION 3

Owner Action Items

The following items require manual action by the project owner. They cannot be applied programmatically or require dashboard access, code commits, or legal review. Each item includes step-by-step remediation instructions.

DEPLOY NOW

Switch AI Provider Import for Production (Code Change Required)

Why it matters: All 5 core service files (rfp-analyzer, knowledge-base, proposal-drafter, compliance-checker, opportunity-discovery) currently import from lib/ai/gemini. Perplexity is the production provider. Until this switch is made, production traffic routes through Gemini.

Remediation steps:

- 1. Create a single provider router at src/lib/ai/provider.ts that re-exports all functions from either lib/ai/perplexity or lib/ai/gemini based on the AI_PROVIDER env var.
- 2. Set AI_PROVIDER=perplexity in Vercel production environment variables.
- 3. Set AI_PROVIDER=gemini in local .env.local for QA testing.
- 4. Update all 5 service file imports from "@lib/ai/gemini" to "@lib/ai/provider".
- 5. Verify PERPLEXITY_API_KEY is set in Vercel production environment.
- 6. Deploy and smoke-test one RFP analysis end-to-end in production mode.

SECURITY

Revoke anon Role EXECUTE on SECURITY DEFINER Functions

Why it matters: Six SECURITY DEFINER functions are callable by unauthenticated users via the Supabase REST API. This means anyone without an account can invoke handle_new_workspace, is_workspace_admin, redeem_workspace_invite, validate_workspace_invite, is_workspace_member, and handle_new_user_workspace_invite. Supabase grants EXECUTE to PUBLIC by default on new functions.

Remediation steps:

- 1. Go to Supabase Dashboard > SQL Editor.
- 2. Run the following SQL: REVOKE EXECUTE ON FUNCTION public.handle_new_workspace() FROM anon, PUBLIC; REVOKE EXECUTE ON FUNCTION public.is_workspace_admin(uuid) FROM anon, PUBLIC; REVOKE EXECUTE ON FUNCTION public.is_workspace_member(uuid) FROM anon, PUBLIC; REVOKE EXECUTE ON FUNCTION public.redeem_workspace_invite(text) FROM anon, PUBLIC; REVOKE EXECUTE ON FUNCTION public.validate_workspace_invite(text, text) FROM anon, PUBLIC; REVOKE EXECUTE ON FUNCTION public.handle_new_user_workspace_invite() FROM anon, PUBLIC;
- 3. Note: handle_new_user_workspace_invite is a trigger function — only reachable via the auth.users trigger, not the REST API. Still revoke as a precaution.
- 4. Verify by attempting to call one function via the Supabase REST API without a JWT — it should return 403.
- 5. Reference: <https://supabase.com/docs/guides/database/database-linter?lint=0028>

SECURITY

Enable Leaked Password Protection in Supabase Auth

Why it matters: Supabase Auth is not checking new passwords against the HaveIBeenPwned.org compromised credentials database. This allows users to register with known breached passwords, increasing

account takeover risk.

Remediation steps:

- 1. Go to Supabase Dashboard > Authentication > Settings.
- 2. Scroll to "Password Security" section.
- 3. Enable "Detect breached passwords" toggle.
- 4. Save changes.
- 5. No code change or redeployment required — this is a dashboard-only configuration.

**BEFORE
PILOT**

Fix Email Confirmation Redirect URL

Why it matters: Supabase sends email confirmation links with a redirect to the Site URL configured in Auth settings. If this is still set to localhost:3000, any user who clicks the confirmation email link on a mobile device or outside the local dev environment will land on a broken page. Must be updated before external pilot users are invited.

Remediation steps:

- 1. Go to Supabase Dashboard > Authentication > URL Configuration.
- 2. Set "Site URL" to: <https://proposal-pilot-vert.vercel.app>
- 3. Under "Redirect URLs", add: <https://proposal-pilot-vert.vercel.app/auth/callback>
- 4. Save changes.
- 5. Test by registering a new user account via the production URL and confirming the email link works end-to-end.

**BEFORE
PILOT**

Add Per-Workspace AI Rate Limiting

Why it matters: The `/api/opportunities/score`, `/api/proposals/[id]/draft`, and `/api/knowledge-base/index` routes have no per-workspace daily call limits. With multiple pilot workspaces running concurrently, a single workspace could exhaust daily API quota or generate unexpected costs. The Agent API is the most expensive call in the system.

Remediation steps:

- 1. Add a `workspace_ai_usage` table (or use `agent_operations` count query) to track daily call counts per workspace.
- 2. In each of the three API routes, check the count of `agent_operations` for the current `workspace_id` where `created_at > today midnight UTC`.
- 3. Return HTTP 429 with a clear message if the limit is exceeded (suggested limits: 50 scoring calls/day, 20 draft calls/day, 10 indexing calls/day per workspace).
- 4. Expose remaining quota in the Computer Ops dashboard for transparency.
- 5. Consider making limits configurable per pilot workspace via the `pilot_workspaces` table.

QUALITY

Improve Knowledge Base Chunking Strategy

Why it matters: The current chunker splits evidence at word count boundaries (500 words, 50-word overlap). This frequently cuts sentences mid-thought and separates paragraph context. For proposal evidence, semantic coherence within a chunk directly determines whether the embedding accurately represents the content — broken chunks produce weaker retrieval matches.

Remediation steps:

- 1. Replace the word-count split in the knowledge base service with a sentence-boundary splitter (e.g., split on ". " or " " then accumulate until ~500 words).
- 2. Keep the 50-word overlap but ensure the overlap includes complete sentences.
- 3. For DOCX files, leverage mammoth's paragraph structure rather than raw text — chunk at natural paragraph boundaries.
- 4. Re-index any existing evidence chunks after the change to rebuild the embedding vectors.
- 5. Test retrieval quality: upload a past proposal and verify that match_evidence_chunks returns contextually relevant passages.

QUALITY

Harden Citation Round-Trip to DOCX Export

Why it matters: The proposal drafter stores citation IDs on each proposal section (citations table, FK to proposal_sections). The docx npm package assembles the export. If the citation reference injection is not verified end-to-end, exported proposals will be missing their evidence traceability — a core product differentiator for government contracting.

Remediation steps:

- 1. Generate a complete proposal draft against a sample RFP using real evidence chunks.
- 2. Export to DOCX and verify that each section includes citation markers pointing back to source document names and page references.
- 3. If citations are missing: review the export function in src/lib/proposals/export.ts and ensure it joins citations.evidence_chunk_id → evidence_chunks.source_document_id → source_documents.filename.
- 4. Add source document name, category, and content_date to each citation footnote in the DOCX.

LEGAL

Legal Review: Success-Fee Model Terms

Why it matters: The success-fee pricing model is a core competitive differentiator but has no drafted contract language, cap, or enforcement mechanism yet. Before exposing this to pilot users, the terms must be defined.

Remediation steps:

- 1. Define the fee structure: percentage of contract value, cap per award, minimum contract size threshold.
- 2. Define the reporting obligation: how does ProposalPilot know when a contract is awarded? (SAM.gov award notice, user self-report, honor system?)
- 3. Draft a simple service agreement addendum covering: fee calculation, reporting window (e.g., 30 days post-award), audit rights, and max fee cap.
- 4. Have a startup attorney (not just AI-generated language) review the fee agreement before pilot users sign.
- 5. Consider whether the success-fee model interacts with any FAR/DFARS regulations on contingency fee arrangements for government contractors.

SECTION 4

Perplexity Billion Dollar Build — Competition Strategy

Competition Status: Active · Submission deadline June 2, 2026 · Finale June 9–10, 2026 · Judges: Aravind Srinivas (Perplexity CEO), Mira Murati (Thinking Machines / ex-OpenAI CTO), + 3 undisclosed late-stage VCs

4.1 How ProposalPilot Fits the Competition

The Billion Dollar Build is judged on four criteria: market size, product quality, user traction, and Perplexity Computer centrality. It is explicitly framed as an investment thesis exercise, not a hackathon — judges are asking "most credible path to \$1B valuation in 36 months."

CRITERION	PROPOSALPILOT POSITION	SCORE
Market Size	\$750B+ annual federal procurement. 500,000+ SAM.gov-registered small businesses. GovDash's \$30M Series B (Jan 2026) independently validates the category. The path to \$1B is arithmetically straightforward: 2% of SMB GovCon market at \$50K ARR = \$500M+ ARR.	STRONG ✓
Product Quality	Live deployed application at proposal-pilot-vert.vercel.app with full pipeline: SAM.gov discovery → RFP analysis → knowledge base → drafting → compliance → export. Load Demo + mock mode functional.	STRONG ✓
User Traction	Currently 1 workspace, minimal data. This is the weakest criterion at present. Need real pilot users with measurable activity before June 2 submission.	NEEDS WORK
Computer Centrality	Currently uses Perplexity Agent API and Sonar deeply — but Perplexity Computer (the Max-tier multi-agent platform) is not yet integrated. This is the single most important criterion. See Section 4.3.	CRITICAL GAP

4.2 The Critical Distinction: Computer vs. Agent API

This is the most important conceptual clarity for the submission. The competition requires **Perplexity Computer** — the Max-tier multi-agent cloud platform — as the "primary and most important AI." Products that use only the Agent API, Sonar API, or Embeddings API will not satisfy the Computer centrality criterion. Judges are looking for Computer being "genuinely core to the product and operations — not bolted on."

	PERPLEXITY AGENT API (Current)	PERPLEXITY COMPUTER (Required)
What it is	Developer API: deterministic, programmatic, single-call async	Multi-agent cloud platform: autonomous, persistent, multi-session workflows
Who operates it	Your backend code calls the API	Computer operates autonomously — spawns sub-agents, makes decisions

Use case	Single RFP analysis, single draft generation, one embedding call	Weeks-long BD pipeline, competitive intelligence gathering, operations
Scale	Scales horizontally via serverless — good for per-proposal tasks	Does NOT scale horizontally — not suited for per-user per-proposal tasks
Competition role	Core product infrastructure — keep as-is	Required for the submission — must be demonstrably central

4.3 Where Computer Belongs in ProposalPilot

Your instinct is exactly right: Computer does not scale well per-user or per-proposal, so it does not belong in the synchronous request-response path of the main product. The correct architecture uses the deterministic Agent API + Sonar for the product's user-facing pipeline, and Computer as the intelligent operator of the business itself. This is actually a stronger story for the judges — Computer is not a feature, it IS the competitive intelligence and operations layer.

The Two-Layer Architecture

LAYER	TECHNOLOGY	ROLE	SCALES?
Product Layer (per user, per proposal)	Perplexity Agent API + Sonar + Embeddings (via pplx-embed-context-v1)	RFP parsing, requirement extraction, drafting, compliance, SAM.gov scoring	YES — serverless
Operations Layer (business intelligence)	Perplexity Computer (Max-tier multi-agent platform)	Competitive intelligence, pilot management, market monitoring, system health, GTM ops	NOT NEEDED — runs daily/weekly

4.4 Specific Computer Integration Opportunities

The following six Computer workflows should be built and demonstrated before the June 2 submission. Each is a real operational workflow that ProposalPilot runs as a business — not a demo feature. Together they show Computer as the operational backbone of the company, which is exactly what top-tier submissions must demonstrate.

1 Competitive Intelligence Monitor (Weekly)

HIGHEST IMPACT

Workflow	Computer runs a weekly multi-agent research workflow: scan GovDash, Unanet, AutogenAI, GovEagle, and Rohirrim for product updates, pricing changes, new funding, and feature launches. Cross-reference with ProposalPilot feature gap analysis. Deliver a structured competitive brief to Supabase (competitive_intel table) + notify via email/Slack.
Judge signal	Directly demonstrates "Computer IS the product operations" — not a search widget. Shows judges a running example of Computer doing weeks of analyst work autonomously.
Tech stack	Computer web search (multi-domain) + Supabase connector + email notification. Runs via cron or Computer's own scheduling.

2 Pilot Workspace Health Monitor (Daily)

HIGHEST IMPACT

Workflow	Computer queries Supabase agent_operations and pipeline_run_metrics daily. Identifies: workspaces with failed pipeline runs, workspaces idle for >3 days, proposal drafts stuck in draft status >7 days, low compliance scores. Generates a prioritized intervention brief and sends to the founder.
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Judge signal	Shows Computer operating ProposalPilot's customer success function autonomously. Directly replaces a CS hire. This is the "3.25 years of work in a month" narrative.
Tech stack	Supabase connector (read agent_operations, pilot_workspaces, proposal_drafts) + Computer analysis + Slack/email output.

3 SAM.gov Market Intelligence Digest (Weekly)

HIGH IMPACT

Workflow	Computer scans SAM.gov award notices for NAICS codes 541512, 541511, 541519. Extracts: winning contractor names, award values, contract vehicles used, agency repeat patterns. Builds a "who is winning what and why" intelligence brief. Stores structured data in a new market_intel Supabase table. Feeds back into the opportunity scoring model as prior award signals.
Judge signal	Demonstrates Computer doing institutional-grade market research that normally requires a BD analyst team. Shows the feedback loop from Computer back into the product.
Tech stack	Computer web search (SAM.gov domain filter) + fetch_url on award notices + Supabase write + structured output.

4 Pilot Customer Outreach & Onboarding (One-time per pilot)

HIGH IMPACT

Workflow	When a new pilot workspace is created (pilot_workspaces table insert), Computer runs a 30-minute onboarding workflow: research the company on SAM.gov and LinkedIn, identify their past awards and NAICS codes, pre-populate their company_profile in Supabase, generate a personalized "here is what we found about your past performance" welcome brief.
Judge signal	Shows Computer delivering immediate value to new users — compressing hours of onboarding into minutes. Judges see traction plus Computer centrality in a single workflow.
Tech stack	Supabase connector (read pilot_workspaces insert trigger) + Computer people/web search + Supabase write (company_profiles) + email output.

5 Proposal Win/Loss Analysis (Post-submission)

MEDIUM IMPACT

Workflow	When a user marks a proposal as won or lost (proposal_outcomes table), Computer runs a debrief: search SAM.gov for the award notice, identify the winning contractor if lost, analyze their proposal strategy based on public past performance, and generate a win/loss lessons brief that feeds back into the workspace's knowledge base.
Judge signal	Closes the learning loop in the product — shows Computer making the product smarter over time. Strong 36-month valuation argument: the longer you use ProposalPilot, the more it learns your win patterns.
Tech stack	Supabase connector (proposal_outcomes) + Computer web search (SAM.gov, USASpending.gov) + knowledge base write-back.

6 Computer Ops Dashboard — Live in Product

SUBMISSION CRITICAL

Workflow	The Computer Ops page already exists in the nav. Populate it with a live feed of all Computer workflow runs: timestamp, workflow name, inputs, outputs, duration, and next scheduled run. This makes Computer's work visible to users — they see the machine working on their behalf. For the submission demo video, this is the proof that Computer is genuinely core.
Judge signal	The Computer Ops page is the single most powerful visual element for the submission video. Judges see Computer running real workflows in a live product with real users — not a demo.
Tech stack	agent_operations table already tracks all AI calls. Add workflow_type and computer_run_id fields. Surface in Computer Ops page with auto-refresh.

4.5 Submission Strategy & \$1B Valuation Thesis

The judges are evaluating ProposalPilot as a VC investment, not a product demo. The 36-month path to \$1B must be arithmetically defensible and memorably framed.

The Math

VARIABLE	VALUE	SOURCE
SAM.gov registered small businesses	500,000+	SAM.gov public data
Realistic addressable (IT services, professional services)	~75,000	NAICS 541xxx filter
Target penetration at scale (5 years)	10% = 7,500 customers	Conservative vs. Deltek's 6,000+
Blended ARR per customer (subscription + success fee)	\$18,000/year	vs. \$12K Deltek entry, \$33/seat Unanet
ARR at 10% penetration	\$135M ARR	SaaS at 7-10x = \$1B+ valuation

The Pitch Narrative (5 Minutes)

Minute 1 — The Problem	Every year, 500,000 small businesses on SAM.gov compete for federal contracts. The average proposal costs \$15,000–\$40,000 in labor to write. Win rates sit below 30%. The tools that exist (Deltek, AutogenAI) cost \$30K+/year — priced for large primes, not small contractors.
Minute 2 — The Product	ProposalPilot automates the entire proposal pipeline: upload your company docs once, point at an RFP, and get a cited, compliance-checked draft in minutes. Powered by the Perplexity Agent API for real-time agency intelligence and your own evidence base for grounded citations.
Minute 3 — Computer as the Operating Brain	Perplexity Computer runs ProposalPilot's operations. Every week, Computer monitors our pilot users' proposal pipelines, scans the competitive landscape, identifies market opportunities, and delivers intelligence that makes the product smarter. Show the Computer Ops page live.
Minute 4 — Traction	[Present pilot workspace data: proposals submitted, RFPs analyzed, requirements tracked, time saved. Show the pipeline_run_metrics and agent_operations counts.] We onboarded [N] pilot users in [X] weeks using Computer for personalized onboarding. [Specific win if available.]
Minute 5 — The 36-Month Path	75,000 addressable IT services contractors on SAM.gov. At 10% penetration and \$18K blended ARR = \$135M ARR. At a 7-10x SaaS multiple = \$1B+ valuation. GovDash validated the category with a \$30M Series B in January 2026 at enterprise pricing. We own the SMB half of their market.

4.6 Traction Sprint (4 Weeks to June 2)

Traction is the weakest criterion right now. The database has 1 workspace and minimal data. The judges explicitly want real users with measurable activity — not a demo. The following sprint plan converts the existing pilot infrastructure into a submittable traction story.

Week 1 (May 4–10)	Build the 6 Computer integrations from Section 4.4. Deploy to production. Populate Computer Ops page. Apply owner action items 1–4.
Week 2 (May 11–17)	Recruit 5 pilot workspaces via warm outreach to IT services contractors you know. Use Computer's onboarding workflow (Integration #4) to personalize their setup. Target: 5 active workspaces each uploading ≥1 company doc and analyzing ≥1 RFP.
Week 3 (May 18–24)	Drive proposal generation activity. Goal: ≥3 proposals submitted per workspace = 15 total submissions. Run the SAM.gov market intelligence digest (Integration #3) and share results with pilot users as a value demonstration.
Week 4 (May 25 – Jun 1)	Collect traction metrics: proposals created, RFPs analyzed, requirements tracked, estimated time saved (agent_operations.duration_ms sum). Record submission demo video showing Computer Ops live feed + active workspace data. Finalize pitch narrative with real numbers.

4.7 T&C; Considerations

Important: The competition T&C; (§8, §9, §10) grant Perplexity a perpetual worldwide license to submission materials and explicitly permit Perplexity to build competing products from your submission. Do not submit your core proprietary training data, knowledge base content, or any SAM.gov integration code you consider a trade secret. Frame the submission around the product experience, architecture decisions, and business model — not proprietary datasets. Consult a startup attorney before the final submission if the IP concern is material to your strategy.

CLOSING

Updated Overall Assessment

ProposalPilot is better positioned than the first brief indicated. The product is live, the database is mature, the architecture is sound, and the market timing is exceptional. Two database migrations were applied today hardening performance and security. The path to a competitive Billion Dollar Build submission is clear: build the six Computer integrations, run the 4-week traction sprint, apply the owner action items, and deliver a pitch grounded in real pilot user data. The \$1B valuation thesis is arithmetically defensible and the competitive gap is real.

ITEM	STATUS	OWNER
Embedding column dimension	Already correct (1024) ✓	—
match_evidence_chunks RPC	Already exists ✓	—
Vercel deployment	Live at proposal-pilot-vert ✓	—
Performance indexes (7 new)	Applied migration 020 ✓	Perplexity Computer
RLS re-evaluation (2 policies)	Applied migration 021 ✓	Perplexity Computer
Mutable search_path (9 functions)	Applied migration 021 ✓	Perplexity Computer
AI provider switch (prod → Perplexity)	Code change needed	Owner — Section 3, Item 1
SECURITY DEFINER anon REVOKE	SQL needed in dashboard	Owner — Section 3, Item 2
Leaked password protection	Dashboard toggle needed	Owner — Section 3, Item 3
Email confirmation URL	Dashboard config needed	Owner — Section 3, Item 4
Per-workspace AI rate limits	Code change needed	Owner — Section 3, Item 5
Computer integrations (6)	Not yet built	Owner — Section 4.4
Traction sprint (5 pilot users)	Not started	Owner — Section 4.6
Success-fee legal review	Not started	Owner — Section 3, Item 8

Prepared by Perplexity Computer · May 3, 2026 · Sources: GitHub/Ryan-Richmond/RFPPProject, Supabase ProposalPilot (fzukcuiwiexrfquuzqii), Vercel (proposal-pilot-vert.vercel.app), Billion Dollar Build: perplexity.ai/hub/blog/introducing-perplexity-computer, x.com/perplexity_ai/status/2041929222135173466, businesstoday.in, idlen.io, reddit.com/r/SaaS, Competitive research: GovDash, Deltek, Unanet, AutogenAI, GovEagle, Rohirrim, Govly